A relation is representable in Q exactly when it is computable

The Open Logic Project contributors

openlogic:incompleteness/representability-in-q:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-incompleteness-representability-in-q/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the forward algorithm and the converse specialisation are both complete in-span

Editorial note. The label thm:representing-rels names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Representable functions compute via proof search. Same unit, \ollabel{thm:representable-iff-comp} (incompleteness introduction section). Gives the converse half: the characteristic function of a computable relation is representable, and its representing formula is the relation's witness.

Source reference. Original theorem thm-representable-iff-comp: sources/olp-incompleteness-representability-in-q/pinned-original-source.tex, source lines 1287–1287. Prerequisite pointer invoked at proof line(s) 1808; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *A relation is representable in* ThQ *if and only if it is computable.*

Proof. For the forwards direction, suppose $R(x_0, \ldots, x_k)$ is represented by the formula $A_R(x_0, \ldots, x_k)$. Here is an algorithm for computing R: on input n_0, …, n_k, simultaneously search for a proof of $A_R(\overline{n_0}, \ldots, \overline{n_k})$ and a proof of $\neg A_R(\overline{n_0}, \ldots, \overline{n_k})$. By our hypothesis, the search is bound to find one or the other; if it is the first, report "yes," and otherwise, report "no."

In the other direction, suppose $R(x_0, \ldots, x_k)$ is computable. By definition, this means that the function $\chi R(x_0, \ldots, x_k)$ is computable. By thm-representable-iff-comp, χR is represented by a formula, say $A_{\chi R}(x_0, \ldots, x_k, y)$.

Let $A_R(x_0, \ldots, x_k)$ be the formula $A_{\chi R}(x_0, \ldots, x_k, \overline{1})$. Then for any n_0, ..., n_k, if $R(n_0, \ldots, n_k)$ is true, then $\chi R(n_0, \ldots, n_k) = 1$, in which case Th$Q$ proves $A_{\chi R}(\overline{n_0}, \ldots, \overline{n_k}, \overline{1})$, and so Th$Q$ proves $A_R(\overline{n_0}, \ldots, \overline{n_k})$. On the other hand, if $R(n_0, \ldots, n_k)$ is false, then $\chi R(n_0, \ldots, n_k) = 0$. This means that Th$Q$ proves

$$\forall y\, (A_{\chi R}(\overline{n_0}, \ldots, \overline{n_k}, y) \rightarrow y = \overline{0}).$$

Since ThQ proves $\overline{0} \neq \overline{1}$, Th$Q$ proves $\neg A_{\chi R}(\overline{n_0}, \ldots, \overline{n_k}, \overline{1})$, and so it proves $\neg A_R(\overline{n_0}, \ldots, \overline{n_k})$. $\square$